

Project Name

Project Goal

Describe the final outcome of the project.

Example: Build an AI-powered surveillance system that can monitor cameras, detect people, recognize faces, and notify users about important events.

Problem & Solution Approach

Problem

What problem exists?

Example: Traditional CCTV systems only record video and require humans to constantly monitor footage.

Solution Approach

How will we solve it?

Example:

1. Capture camera feed.
 2. Process frames using AI.
 3. Detect people and objects.
 4. Recognize faces.
 5. Generate alerts.
 6. Store events.
 7. Show results in a dashboard.
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Features

List everything the system can do.

Example:

- User authentication
- Camera management
- Live monitoring
- Person detection

- Face recognition
 - Unknown face alerts
 - Event history
 - Notifications
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Technical Stack

Example:

Frontend:

- React

Backend:

- Spring Boot

Database:

- PostgreSQL

AI:

- Python
- OpenCV
- YOLO

Authentication:

- JWT
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Architecture (System Design)

Example:

User ↓ Frontend ↓ Backend API ↓ Database

Camera ↓ AI Service ↓ Backend API ↓ Database ↓ Notification Service

Database Design

Users

- id
- username
- email
- password_hash

Cameras

- id
- user_id
- camera_name
- camera_url

Events

- id
- camera_id
- event_type
- timestamp

Alerts

- id
- event_id
- status

API Endpoints

Authentication

POST /register

POST /login

POST /logout

GET /profile

Cameras

POST /cameras

GET /cameras

PUT /cameras/{id}

DELETE /cameras/{id}

Events

GET /events

GET /events/{id}

Alerts

POST /alerts

GET /alerts

PUT /alerts/{id}